Classification and Multi-Source Reconstruction of Extreme Convective Wind Events: Observational Evidence, Mechanism Discrimination, Mixed-Climate Extreme Value Analysis, and Engineering Interfaces

PETW Event Classification and Reconstruction Working Group

Abstract

The reconstruction of extreme convective wind events confronts heterogeneous evidence chains: surface station observations, thunderstorm reports, radar storm objects, polarimetric radar quantities, environmental proxy data, nonstationary time histories and engineering features, and statistical components each describe different physical objects, and their spatiotemporal resolutions, observational denominators, and verification characteristics are not equivalent. This review takes evidence-level separation as its organizing principle, examines the seven inference levels of detection, surface gust classification, parent storm mode, local mechanism, engineering features, statistical components, and downstream interfaces, and delineates at each level the supportable propositions and the inference boundaries that may not be crossed. The existing literature collectively shows that detection does not replace classification, classification does not determine mechanism, parent environment is insufficient to resolve local mechanism, feature extraction does not constitute event truth, and statistical separation is not causal attribution; agreement among sources can improve robustness, but it cannot raise the evidence level of any source. Among the included sources, no single source simultaneously provides independently validated event mechanism truth and complete executable simulation input, and the 50 m model diagnostic wind speed cannot be directly interpreted as an engineering three-second gust. Accordingly, existing evidence is insufficient to establish event labels, Path C ground truth, extreme value analysis sample membership, CM1 validation conclusions, return levels, or design quantities. Reliable linkage from classification evidence to magnitude analysis still requires matched mechanism-resolving observations, explicit denominators and metadata, and executable simulation inputs.

Keywords: extreme convective wind; downburst classification; evidence-level separation; multi-source event reconstruction; mixed-climate extreme value analysis; radar storm object tracking; conditional engineering interface

1 Introduction and literature review

Near-surface destructive convective winds are produced by small-scale, short-duration, strongly nonstationary processes, and the processing paradigms developed for synoptic-scale winds do not transfer directly. The evidence involved in convective wind event reconstruction comes from multiple observing modes, different spatiotemporal resolutions, and heterogeneous verification

environments, forming a multi-layer inference chain that spans meteorological observation, engineering analysis, and statistical modeling. Each piece of evidence has a natural ceiling on the inference level it can reach: surface instrument records and thunderstorm reports possess their own precision and bias structures, radar object tracking and polarimetric radar quantities address different physical questions, nonstationary engineering features depend on the choice of decomposition method, and mixed-climate extreme value analysis is constrained by sample composition and denominator definition. Accordingly, the organizing question of this review is not whether the evidence is sufficient but which inference level each piece of evidence reaches.

Fujita (1985) reported that, after analyzing the Eastern 66 accident of 24 June 1975, he named the reconstructed surface wind system DOWNBURST. Subsequent NIMROD observations prompted him to subdivide downbursts further into microbursts and macrobursts based on the horizontal extent of damaging winds. The 4 km damaging-wind extent is a taxonomic definitional boundary given by that source, not an operationally validated rule or one necessarily resolvable at existing network densities. Terminology alone cannot provide sufficient basis for modern event classification: a single gust record, a pressure or temperature signal, a parent cloud morphology, or a damage report cannot independently establish station-level local mechanism truth.

Approaching the problem from the perspective of storm separation and classification in wind speed records, Gaetano et al. (2014) established an anemometric separation framework for extreme wind events, distinguishing thunderstorm gusts, non-thunderstorm gusts, and intermediate or unclassifiable states. Cook (2023b) advanced the problem to automated multivariate classification, introducing explicit artifact categories and independent upper-tail tests. Chen and Lombardo (2020) applied convolutional neural networks to classify thunderstorm and non-thunderstorm records, using time-series features in place of single-point thresholds. Vallis et al. (2019) implemented cross-regional classification on a South American network, retaining explicit suspicious and unclassifiable states. This lineage persistently retains intermediate or uncertain categories, indicating that non-negligible classification ambiguity remains under existing records, labels, and predictor variables; this phenomenon neither proves by itself that the ambiguity arises from intrinsic properties of storm processes nor can it be simply attributed to a deficiency of any single classifier.

Radar storm object identification and tracking constitutes a separate inference lineage. Johnson et al. (1998) defined the automated identification and tracking of storm cells using a multi-threshold reflectivity method. The verification methodology was systematically set out by Witt et al. (1998), revealing the scale mismatch between radar algorithm predictions and severe weather reports. Subsequent work introduced probabilistic fusion, polarimetric radar precursor characterization, and scan-strategy observability analysis into the object tracking lineage, with each layer answering a different physical question and no layer answering the station-level local mechanism question.

Engineering features and mixed-climate statistics constitute a downstream lineage. Solari et al. (2015) defined a decomposition framework for slowly varying means and residuals, noting that the nonstationary features of convective winds depend on method parameters such as the moving-average period and record length. Romanic et al. (2020) introduced changepoint detection into thunderstorm wind speed records, quantifying the transient character across regions. At the statistical level, Gomes and Vickery (1978) proposed separated treatment of extreme wind speed distributions in mixed wind climates. While increasing descriptive precision, this lineage retains

an unclosed inference link: statistical separation is not causal attribution.

The remainder of this review is organized along the inference levels described above. Section 2 formally defines the evidence levels and the cross-level inferences that are not permitted. Sections 3 to 7 examine, in turn, surface observations and report reliability, the transient character and changepoints of gust time histories, radar storm object identification and tracking, polarimetric radar precursors and downburst candidate evidence, and the separation of parent storm mode and local mechanism. Sections 8 to 10 address multi-source event reconstruction, mixed-climate extreme value analysis, and the conditional interfaces to downstream analysis. In the PETW application context, classification evidence serves only as a conditional input to magnitude analysis, and storm classification continues the existing research lineage. Section 2 accordingly organizes the above inference levels into a level lattice.

2 Evidence levels in event classification and reconstruction

Extreme convective wind events involve seven evidence levels, and the propositions that can be established and the inferences that cannot be drawn at each level have explicit boundaries. The station or instrument observation level can establish the existence and temporal characteristics of wind speed, wind direction, temperature, and pressure records but cannot establish the storm type or physical mechanism to which the record corresponds (Gaetano et al., 2014; Vallis et al., 2019). The report level can establish the time, location, and estimated intensity of an event but cannot eliminate source reliability differences and rounding artifacts. The radar object level can establish the construction, tracking, and probabilistic attributes of reflectivity entities, but the object centroid is not the boundary of hazardous weather (Johnson et al., 1998). The polarimetric radar quantity and environmental proxy levels each describe the in-cloud physical state or the atmospheric environmental conditions within their respective observation windows and spatial scales. The time-history and engineering-feature level yields products of nonstationary decomposition, and feature values depend on the decomposition method. The downstream interface level defines the variables, metadata, and unresolved conversions for cross-level transfer.

The inferential relationships between these levels obey strict constraints. Detection does not imply classification: a single threshold exceedance is a scalar and is insufficient to assign the event to a specific storm family. Classification does not imply mechanism: parent storm mode describes the system-scale organizational form and cannot determine the station-level local mechanism. Parent environment does not imply local mechanism: environments with the same convective available potential energy or wind shear can produce different surface wind fields. Feature extraction does not imply event truth: the time scales and ratios of nonstationary decomposition depend on the chosen mean definition. Inter-model agreement raises the robustness of the estimate but does not raise the evidence level of any contributing source.

Within this framework, insufficient evidence should be reported as an explicit classification conclusion rather than treated as an analysis failure. When the available data cannot satisfy the minimum evidence set for any candidate mechanism label, retaining an undetermined state is more consistent with the inference boundaries than selecting the superficially most likely label. The included literature has not established general thresholds distinguishing candidate, possible, probable, and confirmed confidence states, and therefore this review makes no event classification

determinations but instead delineates the propositions that each type of evidence can and cannot support within its own scope.

Any evidence entering cross-level comparison must carry a minimum set of source information: source identifier, method description, observation cadence, height, averaging period, denominator definition, threshold, missing-data rate, and transfer conditions. Records missing any one of these metadata items remain usable within their own scope but may not be merged directly with records carrying different metadata. The lowest level of the level lattice is the station observation, and Section 3 proceeds from that point.

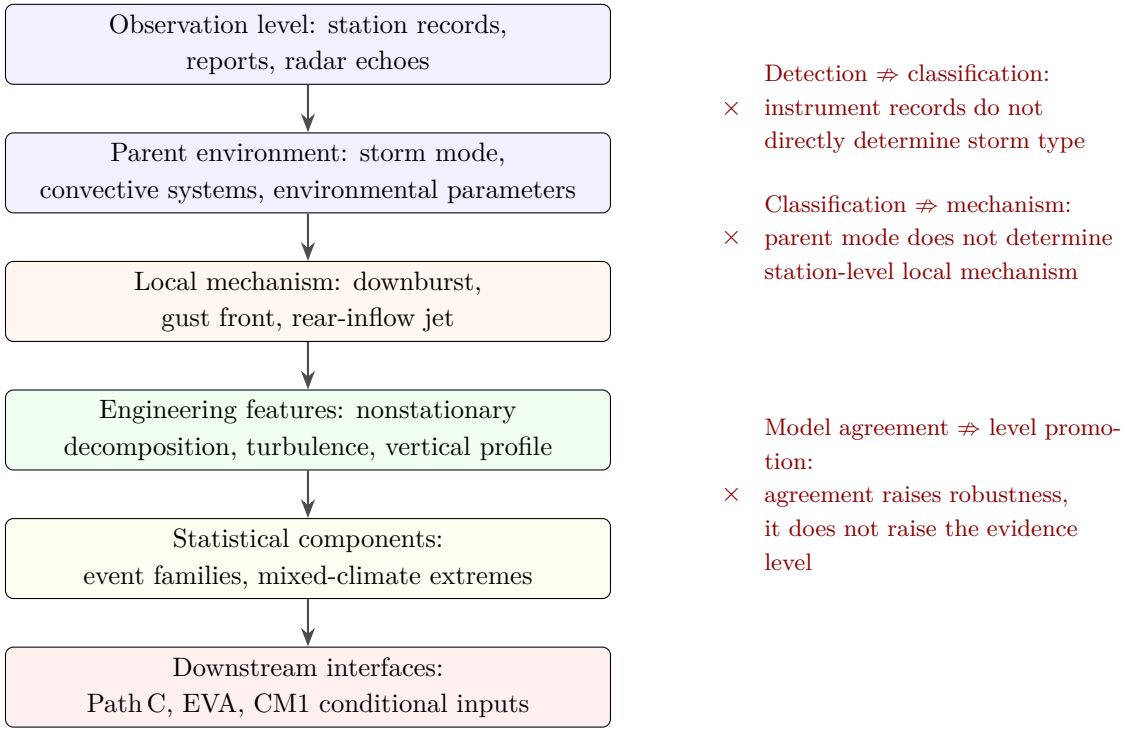


Figure 1: Conceptual synthesis of the evidence levels in event reconstruction and the cross-level inferences that are not permitted. Solid arrows indicate permitted directions of information flow, and the × marks indicate boundaries that inter-level inference may not cross. This figure is a conceptual synthesis of the reviewed literature and contains no numerical results.

3 Surface observing networks, ASOS/Mesonet quality control, and report reliability

Surface observations are the lowest-level input to the event reconstruction chain, but heterogeneous instrument types, observation cadences, exposure classes, and processing histories mean that two records of the same physical quantity are not necessarily comparable. Network density, maintenance status, and real-time quality control algorithms further alter the content of data as it reaches downstream analysis. Consequently, evidence existence and evidence reliability are two questions that must be handled separately.

Gunter and Long (2024) provided a convective gust climatology from the West Texas Mesonet. The network contains 33 stations, the observation period runs from 1 January 2005 to 31 De-

cember 2019, the typical inter-station spacing is about 35 km, sensor height is 10 m, and the record structure consists of independent 5 minute intervals, each containing the mean of the 3 s sampling speed, the vector-mean direction, and the peak 3 s gust. That source recorded 1,042 radar-associated severe convective gusts, with the severe gust threshold at 25.7 m/s or greater and the significant severe gust threshold at 33.4 m/s or greater. The mean monthly data availability is 99.9%. The direction variable is the concurrent 5 minute vector-mean direction, and the direction of the peak gust is not recorded. The radar association establishes convective or outflow association but does not specify storm mode.

The dry/wet classification in that source gives internally contradictory values at different locations. By individual gust statistics, the page 15 text reports 55.9% as dry type, while Figure 13 on the same page shows 642 dry and 400 wet out of 1,042 gusts, corresponding to 61.6% and 38.4%, and the page 16 text separately reports 38% as wet type (Gunter and Long, 2024). The median gust speeds of dry-type and wet-type individual gusts are 27.1 m/s and 28.0 m/s, respectively, and are statistically different at the 95% confidence level. By station-specific event statistics, 651 station events are identified under a separation rule of greater than 30 minutes; the page 15 text reports that 12% of events contain at least three consecutive severe gusts, while the pie segments of Figure 14 on page 16 sum to 11% after rounding. The event-scale wet-type fraction is 45% in the page 16 text, and Figure 15 shows 298 of 651, that is 45.8%. The median event speeds of dry-type and wet-type events are 27.1 m/s and 28.5 m/s, respectively, and are statistically different at the 95% confidence level. These branches are not reconciled within the source and are presented side by side as given.

Under the figure-consistent after-minus-before sign interpretation, the thermodynamic perturbation medians are temperature $-6.2\,^{\circ}\mathrm{C}$, dewpoint $-0.06\,^{\circ}\mathrm{C}$, relative humidity +22.2 percentage points, and pressure +1.4 hPa (Gunter and Long, 2024). That source reports that 50.85% of events show a dewpoint decrease; the exact decrease and increase counts are not reported. The increase in relative humidity is directly supported, but an increase in absolute moisture is not confirmed. The sign definition conflicts between page 6, which states after minus before, and page 16, where the text implies before minus after; the figures and figure captions are consistent with after minus before, and this conflict is retained as unresolved. Among the 76 positive-temperature-perturbation events, 57 also show a dewpoint decrease. Among the 11 events exceeding a borrowed warming threshold, 9 also show drying; these are treated as suspected heat burst candidates identified with an approximate threshold. That source notes that radar classification would be required to distinguish heat bursts from rapid boundary-layer recovery, and a confirmed heat burst occurrence rate cannot be established from the available data.

Cook (2023a) analyzed the impact of ASOS real-time quality control on convective gust extremes. The verified authoritative version of that source is the record on EngrXiv labelled Preprint / Version 1; Crossref classifies it as posted content of subtype preprint, and the available material is insufficient to establish its equivalence with a journal version. The analysis scope covers 860 CONUS ASOS stations screened to 450 with WMO exposure class 1 or 2, using gust events exceeding 20 kn during 2000 to 2021, limited to Class 4 and Class 5. Test 10 triggers an alarm when the running two-minute mean is at or below 6 kn, the running 3 s mean exceeds the applicable gust threshold, and the gust exceeds 2.5 times the two-minute mean; upon triggering, speed and direction records are suppressed for 5 minutes. The gust threshold was changed from greater than 6 kn to greater than 13 kn at the end of 2017.

A source-internal conflict exists regarding the denominator of peak loss: the page 1 abstract phrases it as about 10% of these events, while the page 6 body and the page 14 conclusion limit the about-10% figure to simulated Test 10 false alarms (Cook, 2023a). The usable formulation is the alarm-limited version; the unqualified 10% of events formulation is prohibited, and this wording conflict is retained. About 10% of stations show a discernible simulated 50-year impact. The largest Class 4 underestimate is station KBKB at the 6 kn threshold, 4.3 kn or 10.7%, with a rank drop from 1 to 22, and about 1% high at the 13 kn threshold. The largest Class 5 underestimate is station KTQE, 5.2 kn or 5.7% at both thresholds. Pre-2014 affected locations and observed rejection rates do not predict the 2014 to 2021 underestimate locations. The 3-degree neighbour lower bound proposed by that source clips about 40% of stations, including legitimately low stations, and is explicitly labelled preliminary.

Report reliability constitutes an independent evidence level. Edwards et al. (2018) quantified the reliability structure of measured versus estimated values in convective wind reports: across ten years and 150,423 convective wind reports, 15,183 are measured and 135,240 are estimated or sustained. The 50 kn estimate cluster of 54,229 reports against 366 reports at 51 kn is an order-of-magnitude rounding artifact that reflects rounding habit rather than a physical distribution feature. The estimated-to-measured ratio differences between states are as large as 78:1, reflecting jurisdiction and network coverage differences rather than meteorological gradients. High-resolution dual-Doppler observations (Gunter and Schroeder, 2015) recorded positive biases of 2.2 m/s and 2.37 m/s at 10 m, with bias decreasing with height, confirming that radar wind speed below 50 m is not an unbiased surface gust. Lombardo and Zickar (2019) provided characteristics of measured extreme thunderstorm near-surface gusts from a national network. Iwashita and Kobayashi (2019) recorded meteorological variable transitions during downburst occurrence using a high-density surface network, but the network-specific transition thresholds are not universal diagnostic standards. Automated classification methods (Cook, 2023b; Chen and Lombardo, 2020; Vallis et al., 2019) provide classification-level evidence within their respective networks and climate conditions, but classifier output is not a confirmed event label or mechanism truth. The station and report levels provide evidence carrying quality labels rather than generalizable conclusions, and a positive ASOS or Mesonet record is not Path C ground truth. Station records are inherently time series, and their transient structure is the subject of the next section.

4 Transient character, changepoints, and detectability of gust time histories

A threshold exceedance is a scalar, and the evidence that distinguishes event families resides in the internal structure of wind speed time series. Peak duration, rise rate, the locations of abrupt changes in the mean and standard deviation, and post-decomposition ratios from nonstationary analysis can describe the transient features of an event, but these descriptors depend on the chosen decomposition method and the record conditions.

Romanic et al. (2020) performed changepoint analysis on 19 thunderstorm wind events from Europe, the United States, and Australia. The abstract states that the sample contains 41 wind speed records, while the body states 42; this internal count discrepancy is retained. The method

uses a penalized-likelihood optimal partitioning framework, applying a mean cost function that includes a piecewise linear trend and a standard-deviation cost function with the trend coefficient set to zero. The slowly varying mean is prefiltered with a 30 s moving average. Two of the records are digitized from published time-series figures and carry larger uncertainty than the original measurements. Most station roughness values are ESDU category estimates rather than measurements.

Under the mean statistic, about 60% of records have a peak duration of 2 to 5 min; under the standard-deviation statistic, about 60% fall in 5 to 10 min, and no record is below 2 min (Romanic et al., 2020). The proportion of rise times below 2 min under the mean statistic is reported as about 60% in the Results section and as about 50% in the Conclusions section; both branches are retained side by side. The proportion of rise times below 1 min under the standard-deviation statistic is less than 10%. European acceleration ratios are typically 2 to 4, and the mean-based acceleration ratio is about 1.65 times the fluctuation-based ratio; the authors state that the US sample is too small to support a regional comparison.

A changepoint is a statistical partition of a wind speed record, not mechanism identification. The storm mode labels attached to the analyzed events are inherited from other sources and are not findings of the changepoint method. In intermediate-intensity wind speed records, the absence of mean and standard-deviation changepoints can serve as a candidate discriminator for describing intermediate categories, but this null result has not been independently validated as a classification criterion in an independent sample, so it may not be credited with established classification performance. Bowen (1996) provided a single separately recorded outflow event with a jet-like vertical structure and reduced turbulence intensity during its passage, constituting feature-level evidence.

Transient character and detectability are directly linked. The response limitations of cup anemometers produce about 13% attenuation at 30 m wavelength and about 71% attenuation at 17 m wavelength for a distance constant of 2.7 m, and part of the US sample shows spectral behavior steeper than the inertial subrange; this anomaly is not explained in the source (Romanic et al., 2020). The penalty parameter depends on record length and sampling frequency and may not be cited as a universal default. Gunter (2015) provided outflow time history and profile evidence using mobile radar and rapidly deployed surface observations, Orwig and Schroeder (2007) quantified the near-surface characteristics of extreme thunderstorm outflows, and He et al. (2024) obtained independent observations of downburst wind characteristics through Doppler profiler and near-surface measurements. These feature-level results carry their respective instrument and site conditions and do not constitute universal design inputs. Surface time series cannot resolve parent objects, and radar storm object algorithms are designed precisely to construct parent objects.

5 Radar storm object identification, tracking, and algorithm verification

A storm object is a constructed entity, not an observed entity. The three-dimensional voxel envelope produced by applying multi-threshold segmentation, connectivity analysis, and vertical association to radar reflectivity fields defines an algorithmic object whose center, boundary, and

lifetime depend on construction parameters rather than on a unique physical boundary.

Johnson et al. (1998) defined the Storm Cell Identification and Tracking (SCIT) algorithm. That algorithm processes reflectivity only and does not process radial velocity, and it uses seven thresholds (60, 55, 50, 45, 40, 35, and 30 dBZ), a default segment length of 1.9 km, a component area of 10 km^2, an azimuthal separation of 1.5 degrees, a segment overlap of 2 km, and a minimum of two segments. The vertical association search radii are successively 5, 7.5, and 10 km. Temporal association is based on a first-guess position, and association terminates when the data interval exceeds 20 min. The motion vector is derived from an equally weighted linear least-squares fit to the current and up to ten previous positions.

On the same manually verified dataset of 17 selected cases and 6,561 cells, with all algorithms at their default settings, the SCIT probability of detection is 28%, 68%, and 96% in the 30 to 39 dBZ, 40 to 49 dBZ, and at-least-50 dBZ intervals, respectively, while the older storm series algorithm gives 17%, 24%, and 41% (Johnson et al., 1998). Across four tracking cases, the correct temporal association rate is 91% to 97%. Position forecast error grows from 2.0 km at 5 min to 22.8 km at 60 min, and the SCIT tracked sample decays from 898 cells having at least two volume scans to 55 cells at 60 min. The 96% figure is not an all-storm detection rate but the value for the at-least-50 dBZ interval in selected cases. The 898-cell forecast sample was not independently manually verified. The stratiform case gives 0% in the 30 to 39 dBZ interval and 13% in the 40 to 49 dBZ interval, and the at-least-50 dBZ value is not reported; therefore the absolute statement that the algorithm cannot detect stratiform precipitation echoes is incorrect. The 90-degree directional gate was demonstrated but not implemented, and it is a configuration note. The centroid of an object is not the boundary of hazardous weather or the footprint of a threat area.

Witt et al. (1998) revealed the scale mismatch between county-level warning-duration verification reports and storm cell and radar volume-scan algorithm output. In one specific case, a supercell lasting 6 hours and crossing six counties requires six reports to complete warning verification, while 5 min-interval volume scans produce 78 algorithm predictions, leaving 72 unverified. About 30% of hail reports do not correlate well with cell positions. Tornado report time error is at least 5 min in 38% of cases and position error is at least 5 km in 12% of cases, with 12% of time errors exceeding 40 min. The recommended general scoring window runs from 15 min before report start time to 5 min after report end time. The absence of null observations and the resulting false-alarm inflation risk constitute an unresolved structural verification problem. Algorithm flags are not local wind mechanism truth, and the recommended scoring window is not a universal event association rule.

At the probabilistic fusion level, Gard et al. (2022) analyzed the applicability of ProbSevere v2 to pulse thunderstorms, identifying 365 severe storm objects and 77,800 null objects in one regional season, with retrospective centroid-to-report matching within about 0.1 degrees and a 2 min window and a minimum object lifetime of 8 min. The median probabilities near the severe time are 11% and 10%, and 47.40% of severe objects remain in the lowest probability bin throughout. Cintineo et al. (2024) advanced data fusion and machine learning to ProbSevere v3, adding fusible input layers. Fused storm objects undergo merging and splitting, and consequently the identity, lifetime, and area of an object need not correspond to a single physical storm cell. Retrospective centroid-to-report matching is not mechanism ground truth. Object algorithms describe where storms are located; polarimetric radar quantities attempt to describe what physical

processes are occurring inside storms.

6 Polarimetric radar precursors and multi-source downburst candidate evidence

Polarimetric radar provides microphysical and kinematic information beyond reflectivity, including differential reflectivity, specific differential phase, and correlation coefficient. A polarimetric quantity anomaly that is associated in space and time with strong surface winds can serve as a precursor candidate, but a precursor signature must still be distinguished from a mechanism: the presence of a signature does not determine the nature of the mechanism, and the absence of a non-downburst control group in the source means that specificity and false alarm rate cannot be established.

Gibson and Carlin (2025) performed polarimetric radar precursor analysis based on automated tracking for 53 reported wet downburst cases. These cases are distributed over 38 days, at distances of 30 to 100 km from the nearest radar, with a regional distribution of Northeast 8, Midwest 14, Southwest 19, Southeast 12, and Northwest 0. That study uses a VIL-based watershed storm object algorithm, repairs object identity for 13 merged cases through a manual common identifier, applies vertical interpolation to a 50 m grid, defines low-level divergence at the 0.5-degree elevation, and evaluates midlevel convergence from 3 to 7 km. Area-integrated divergence is defined as the sum, over gates whose divergence exceeds 0.0025 s^{-1}, of the divergence multiplied by the gate sector area; this is explicitly not a sum of area alone. Two distinct 1 km thick specific-differential-phase integration layers are centered on the freezing level and on 1 km below the freezing level, each extending about 500 m above and below, and are integrated for gates exceeding 1 degree/km. The manual anchoring window runs from 20 min before to 10 min after the divergence signature time, where that time is defined as the first scan reaching the divergence threshold or the first manually identified appearance of divergent inflow and outflow velocities.

Within the signature window, 44 of 53 cases contain a descending reflectivity core, 45 contain a descending specific-differential-phase core with an additional 4 marginal non-descending cores, 47 show a differential reflectivity trough with a collocated correlation coefficient decrease, 44 reach the midlevel convergence threshold, and 36 reach the low-level divergence threshold (Gibson and Carlin, 2025). These within-window fractions are not interchangeable with the presence or threshold fractions separately tabulated in the distance table of that source: the row time ranges, denominators, and missing-data handling of the distance table are not disclosed. An unresolved source-internal conflict exists between the joint within-window value of 47/53, that is 0.89, and the separately tabulated values of 0.91 for the differential reflectivity trough and 0.85 for the collocated correlation coefficient decrease: because 0.85 is below the joint 0.89, a simple superset interpretation is impossible.

Within 65 km of the storm report location, the low-level divergence occurrence rate is 0.84, falling to 0.52 beyond 65 km, and the occurrence rate of upper-level signatures is far less sensitive to range than that of low-level divergence (Gibson and Carlin, 2025). The composite timing sequence shows that the freezing-level phase quantity appears 10 to 20 min before the anchor time, the layer 1 km below appears 5 to 10 min before, differential reflectivity column area and

volume appear about 15 min before, and midlevel convergence and vertically integrated liquid appear at or immediately before the anchor time.

The case set does not include a non-downburst control group, and therefore specificity, false alarm rate, operational lead time, and out-of-sample skill are not established. Signatures interpreted as precursors may occur in non-downburst storms. The maximum range is too permissive for low-level velocity coverage, and low-level divergence can occur below the radar beam. Under a case-control design, Amiot et al. (2019) used 32 threshold cases and 32 screened null cases at one site during one warm season for C-band polarimetric signature analysis; the direct precipitation-ice-height criterion gives a detection rate of 0.69, a false alarm ratio of 0.12, and a mean lead time of 20.5 min, the gradient criterion gives a detection rate of 1.00 but a false alarm ratio of 0.43, and the indirect criterion gives a mean lead time of 81.6 min arising from early multicell cycling and the fixed-site arrival environment; the screened-null design removes population meaning. Pearson et al. (2023) evaluated the impact of different scan strategies on precursor observability in one simulated case, with update intervals ranging from about 4.5 min to 15 s; rapid imaging is accompanied by beam broadening and sidelobe contamination, a single 10 m/s maximum radial differential velocity criterion can occur throughout the analysis period and therefore divergence area is needed to isolate the ground-contact interval; the best surface divergence error is 3.05 m/s and 6.67 km^2, and freezing-level reflectivity and phase enhancement appear about 7 to 8 min before ground arrival. The precursor evidence, case-control signature evidence, and single-simulation observability evidence together define a candidate signature set, not a validated detector. The precursor and object evidence describes parent storms, and from this point the argument returns to the separation of mode and mechanism.

7 Separation of parent storm mode, local mechanism, and ambiguous states

The radar-object-based taxonomy of parent storms organizes system environments into recognizable storm modes. Sui et al. (2026) used gridded three-dimensional radar reflectivity data (GridRad) for the warm seasons of 1995 to 2017 to analyze more than 40,000 warm-season storms in four Texas cities, defining objects by a 40 dBZ high-reflectivity seed with 20 dBZ precipitation-volume expansion and classifying storms into five families: cellular, isolated, tropical, warm-frontal, and cold-frontal. Smith et al. (2012) performed classification and climatology of convective modes for the contiguous United States. Both sources show that parent storm mode describes system-scale organization, and the thresholds and classification rules adopted are conditioned on each source's observation window and region and require regional recalibration before use in other climates and networks. Parent storm mode is not station-level local mechanism truth, and the results of Sui et al. (2026) may not be used as Path C ground truth.

The system environment level includes parent system labels such as squall lines (Johns and Hirt, 1987), derechos (Ashley and Mote, 2005), and mesoscale convective systems, whose dynamics are governed by cold pool outflow, environmental wind shear, available potential energy, and system-scale circulation (Rotunno et al., 1988). The generation and maintenance of the rear-inflow jet require system and environmental evidence (Weisman, 1992) and cannot be inferred from surface gust speed alone. These labels do not prove station-level local mechanism.

The local mechanism level requires near-surface evidence of destructive divergent outflow together with explicit spatial-scale tracking. A downburst label requires ground evidence of divergent outflow, and the subdivision into microbursts and macrobursts uses the 4 km damaging-wind extent as the boundary (Fujita, 1985); this boundary is a taxonomic definition given by that source and is not an operationally validated rule or one necessarily resolvable at existing network densities. The gust front life cycle involves multi-signal surface time series (Wakimoto, 1982), and a gust front label does not exclude an embedded downburst or rear-inflow jet. Proctor (1988) and Proctor (1989) described, through idealized simulations, the dynamic structure and configuration sensitivity of isolated microbursts, respectively, constituting exploratory physical background.

An ambiguous state is a legitimate classification result. The classification method lineage persistently retains intermediate or unclassifiable categories (Gaetano et al., 2014; Vallis et al., 2019); the absence of mean and standard-deviation changepoints in gust time histories can serve as a candidate discriminator for describing intermediate categories, but its classification performance has not been independently validated (Romanic et al., 2020). Brown et al. (2023) identified multi-source severe convective wind event archetypes in eastern Australia that distinguish system types from local mechanisms, but these archetypes are not cross-regionally transferable labels.

The overall framework of separation rules can be summarized as follows: parent system labels, local mechanism labels, the Fujita scale taxonomy, and the changepoint null results from Section 4 constitute four non-interchangeable layers. The cross-level inferences that the existing sources cannot support include the direct determination of local mechanism from parent mode, the confirmation of event mechanism from a single source, the substitution of environmental clustering for local mechanism evidence, and the equating of classification output with event truth. Schumacher et al. (2023) showed in a simulation of a destructive macroburst that gravity wave breaking can produce strong surface winds, but confirming that mechanism pathway requires multi-layer reconstruction rather than the direct transfer of simulation results. The separated layers must be recombined in event reconstruction, and this constitutes the problem of the next section.

8 Multi-source event reconstruction, confidence propagation, and exclusion logic

The goal of multi-source event reconstruction is to organize evidence from different levels into a multi-layer description of a single event while maintaining the independent source information and denominator definitions of each level. Reconstruction faces two failure modes: equating detection with classification, and equating parent morphology with local mechanism.

In individual event descriptions, station records, reports, radar objects, polarimetric radar quantities, environmental proxy data, time history features, and engineering descriptors constitute evidence levels that can be juxtaposed but not merged. Childs et al. (2021) demonstrated the possibility of multi-source combination through a high-resolution observational reconstruction of a destructive macroburst, but a single reconstruction case is not generalizable. Borchardt et al. (2024) combined radar signatures with matched surface observations to analyze destructive surface winds produced by elevated convection, constituting a mechanism candidate still awaiting multi-layer evidence testing.

Inter-layer agreement raises the robustness of the estimate, not the evidence level. Incompatible denominators must be retained: the network denominator is a station-specific event count, the radar denominator is a volume-scan-level detection probability, the precursor denominator is a within-window case count, and the report denominator is a mixture of estimates and measurements. Romanic et al. (2022) quantified the separation capability of the environmental proxy level for downburst events, but environmental separability does not constitute event-level mechanism confirmation. Hadavi and Romanic (2024a) and Hadavi and Romanic (2024b) used machine learning environmental clustering and downburst precursor parameters, respectively, as proxy-level evidence, and neither can transfer across climates.

Exclusion logic is based on source-specific insufficiency rather than a generic checklist. A single station record, a single report, a damage report, a single radar object, a classifier output, an environmental cluster, or a fitted distribution is individually insufficient to establish local mechanism truth. Satellite environment and lightning environment have no citable source in the included corpus providing substantive synthesis and are therefore retained as open evidence elements. Reconstruction output enters statistical analysis, where separation decisions have tail consequences.

9 Event family separation, mixed-climate extreme value analysis, and statistical boundaries

When records and causal evidence permit, the separation of event families or mechanism candidates is a conditional preference for mixed-climate extreme value analysis (Gomes and Vickery, 1978; Cook, 2004). This preference is conditional: not every site has separable mechanisms, and direct composite fitting is not necessarily invalid when evidence is insufficient. Candidate events may enter a corresponding statistical component only after mechanism-family evidence satisfies predefined stratification conditions; when this evidence is missing, membership should remain undetermined.

The parent wind speed distribution can serve as a theoretical support proxy for the wind speed vector marginal (Harris and Cook, 2014); a successful marginal fit neither proves a single physical mechanism nor recovers event-family structure. Distribution fitting is a statistical operation, not a mechanism classifier.

In suspected mixed climates, the conditional risk of unseparated analysis manifests as the possibility that the tail shape and return level estimates of extreme value distributions may be affected by unseparated mechanism families (Trapp et al., 2010). Fan and Pang (2019) provided stochastic track modeling of tornado hazard, constituting adjacent but not equivalent non-mechanistic statistical evidence. Mohr et al. (2017) showed through screening methods for German convective gust statistics that cross-network processing differences can alter sample composition. These interpretations remain preliminary in the included literature and therefore require explicit conditional wording.

Quality-control-driven upper-tail loss, station event rates, and denominator definitions alter the sample before any distribution choice is made (Cook, 2023a; Gunter and Long, 2024). Cook (2024) linked classification inputs with tail statistics, demonstrating a possible pathway for category-conditional treatment. Zhang et al. (2018b) provided mixed-climate extreme distribu-

tions based on high-frequency observations at port sites, and Lombardo et al. (2021) provided an uncertainty structure for prediction workflows of non-synoptic-scale winds. The regional calibration conditions of these sources differ, and results cannot transfer across networks without recalibration. Statistical readiness is one of several downstream interfaces.

10 Conditional interfaces to Path C, EVA, CM1, and wind engineering

The transition from classification evidence to magnitude analysis requires the definition of interfaces: which variables must cross the interface, in what form, carrying what metadata, and under what unresolved conditions. An interface is characterized by its variables and unresolved conversions.

The engineering feature interface requires that metadata for the nonstationary decomposition method, the moving-average period, duration, turbulence intensity and integral scale, vertical profile, and velocity ratios be transmitted together with each descriptor (Solari et al., 2015; Zhang et al., 2018a). Canepa et al. (2020) reported that nose-shaped profiles occur mainly during the rising and peak phases, that conventional ten-minute averaging can filter out these transient features, and that during the studied downburst direction was approximately height-invariant. Thus, conventional long-period averaging can remove these profile transients, and feature values are stored separately from mechanism labels.

Moore (2024) reported 97 idealized simulations using CM1 Release 20.3, with a horizontal domain of 80 km × 80 km, horizontal spacing of 0.1 km, depth of 15 km, 84 vertical levels, vertical spacing of 0.1 km below 2 km, the lowest velocity level at 50 m AGL, a time step of 1 s, an output interval of 60 s, and an integration time of 2 h. The initiation condition is a single 2 K warm bubble centered at 0.5 km at the domain center, 10 km wide and 1.5 km deep, fixed across the entire ensemble. The microphysics scheme is the Morrison double-moment scheme, the subgrid treatment is the Deardorff scheme, the lower boundary condition is semi-slip, and the lateral boundaries are open radiative. The sounding selection path proceeds from 49,181 wind damage local storm reports, screened environmentally to 13,069, and then screened for moisture to 12,539 candidates. The source-specific results include the following: about 35% of simulations reach or exceed the 26 m/s severe threshold; the mean non-contiguous duration of severe events is 5.2 min with a range of 1 to 17 min, and about 25 min above 20 m/s; the median severe coverage area is of order 10^0 km^2; and the ensemble maximum is 37 m/s. Synthetic network sampling probabilities are 43%, 18%, and 0.7% for networks of mean spacing 2.3 km, 5.2 km, and 33.4 km, respectively. The mean of the 50 m to 10 m maximum wind speed ratio is 1.97. The missing reproduction inputs reported by that source include the following: the exact namelist and the candidate sounding dataset are available only on author request, and the time integration scheme, acoustic treatment, Coriolis treatment, and build and run provenance are not reported. All of the above values are source-specific idealized results and are not a PETW reproduction or validation. The 50 m vortex diagnostic wind speed is not a measured or validated engineering three-second gust. The 1.97 ratio is a model-internal diagnostic and not an engineering conversion coefficient. The synthetic network sampling probabilities are conditioned on an idealized vortex, a 50 m diagnostic, a selected severe subset, and random or regular station placement.

Existing evidence is insufficient to determine EVA sample membership or Path C ground truth directly from event classification. Classification candidates entering extreme value analysis screening still require the event-family evidence described above, and the cross-regional verification of this evidence chain in the included literature remains incomplete. In the PETW application context, the linkage from classification evidence to magnitude analysis can exist only as a conditional interface whose availability depends on the completeness of variables, metadata, and independent verification information on both sides of the interface. Romanic (2024) proposed deterministic and constrained stochastic outflow models, Kwon and Kareem (2019) proposed a gust front factor framework oriented toward codification, Wu et al. (2026) discussed the impact of climate change on extreme windstorms and its implications for structural design, and these sources provide downstream interface options or environmental background rather than design values or code recommendations. The laboratory experiments of Romanic and Hangan (2020) explored the interaction between atmospheric boundary layer winds and downburst outflows under scaled conditions, and the idealized simulations of Proctor (1988) and Proctor (1989) described the physical structure and configuration sensitivity of microbursts; all of these constitute exploratory physical background.

The remaining interface requirements include the gust conversion definition from the model 50 m level to the engineering 10 m reference height, for which no source in the included literature provides a value; the consistency of roughness and friction treatment; averaging and filtering metadata for vertical profiles; and the effect of recording cadence on sampling probability. The interfaces expose the most fragile positions in the inference chain, and this constitutes the subject of the discussion.

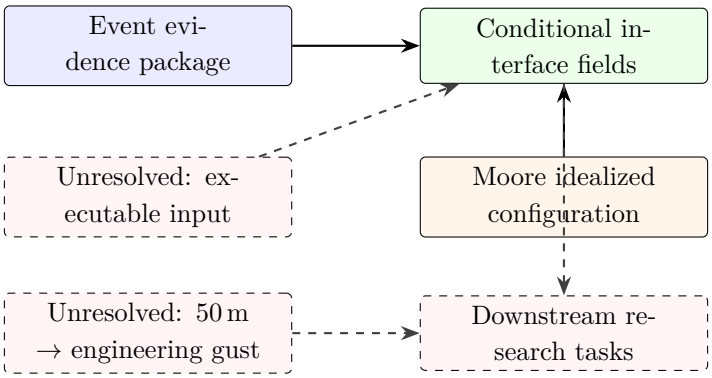


Figure 2: Conceptual synthesis of the information flow from the Event evidence package to the conditional downstream interfaces. Solid arrows indicate defined conditional inputs, and dashed arrows indicate unresolved conversions or inputs. The Moore configuration is an idealized result specific to that source. This figure contains no validation marks, no completion-benchmark symbols, and no design quantities.

11 Discussion

Five categories of high-risk transitions exist in the inference chain: from detection results to event labels, from parent environment to local mechanism, from feature quantities to event truth, from statistical fits to causal interpretation, and from source metadata or evidence type directly to

scientific propositions. The included literature displays the corresponding distortion pathways, including treating a positive surface record as event truth, treating a radar object as a physical storm, treating classifier output as a mechanism label, treating fitted parameters as mechanism evidence, and treating a preprint record as a verified journal version. Restricting these transitions does not narrow the research scope; it keeps propositions at the same inference level as their evidence.

Transferability across networks, climates, instruments, heights, and resolutions is not established in the included literature for classifiers, radar thresholds, environmental clustering, or engineering thresholds. The classification rules of the West Texas warm season do not automatically apply to other regions, the within-window fractions of polarimetric radar precursors cannot transfer to different radar and climate conditions, the penalty parameters of changepoint methods depend on record length and sampling frequency, and the spatial and duration results of idealized simulations are conditioned on their specific configurations.

The asymmetry of denominators and observability is the deepest reproducibility problem. The denominator of surface quality control is a station-specific event count or simulated alarm count, the radar detection probability is denominated by reflectivity interval, the verification scale mismatch is denominated by county-level warnings while algorithm output is denominated by volume scans, and the synthetic network sampling probability is denominated by idealized vortices. These denominators are not interchangeable, and their differences directly affect the effectiveness of confidence propagation. When multi-layer evidence is pooled, the denominator difference is itself information; ignoring this difference widens the scope of the proposition.

12 Concluding remarks

Through evidence-level separation, this review has distinguished the inference boundaries between detection, classification, parent mode, local mechanism, engineering features, statistical components, and downstream interfaces. The fully verified West Texas Mesonet and ASOS real-time quality control studies revealed three categories of key uncertainty in the surface evidence: the multi-branch dry/wet classification conflict, the thermodynamic perturbation sign-convention conflict, and the peak-loss denominator conflict. Changepoint analysis quantified the transient character of thunderstorm gusts while demonstrating that statistical partitioning cannot carry the function of mechanism identification. Radar object tracking provided detection boundaries conditioned on reflectivity intervals and selected cases, and polarimetric radar precursor research cannot establish specificity because it lacks a non-downburst control group. The conditional interface that results frames the linkage between classification evidence and magnitude analysis explicitly as a combined problem of variables, metadata, and unresolved conversions.

Five interrelated research gaps remain in the included literature. One candidate source in the predefined corpus cannot enter substantive synthesis because its authoritative full text is unavailable. Classifiers, radar thresholds, environmental clustering, and engineering thresholds lack cross-climate and cross-network transfer testing. Precursor thresholds and taxonomic scale boundaries lack verification that includes null samples. The existing corpus does not provide station-matched, mechanism-resolved, event-level ground truth. The Event to CM1 interface still lacks complete executable input and a clearly defined, independently verified engineering

gust conversion. Satellite environment and lightning environment are likewise retained as open evidence elements. Existing evidence cannot yield event labels, Path C ground truth, extreme value analysis sample membership, CM1 benchmark selection, return levels, or design values. Closing these gaps requires matched multi-source event reconstruction, threshold verification datasets that include null samples, and clearly defined and independently verified conversions from model diagnostics to engineering gusts.

References

Corey G. Amiot, Lawrence D. Carey, William P. Roeder, Todd M. McNamara, and Richard J. Blakeslee. C-band dual-polarization radar signatures of wet downbursts around cape canaveral, florida. *Weather and Forecasting*, 2019. doi: 10.1175/waf-d-18-0081.1. URL https://doi.org/10.1175/waf-d-18-0081.1.

Walker S. Ashley and Thomas L. Mote. Derecho hazards in the united states. 2005. doi: 10.1175/BAMS-86-11-1577. URL https://journals.ametsoc.org/view/journals/bams/86/11/bams-86-11-1577.xml.

Brett S. Borchardt, Keith D. Sherburn, and Russ S. Schumacher. Radar signatures and surface observations of elevated convection associated with damaging surface winds. 2024. doi: 10.1175/WAF-D-23-0171.1. URL https://journals.ametsoc.org/view/journals/wefo/39/6/WAF-D-23-0171.1.xml.

Brent M. Bowen. Example of reduced turbulence during thunderstorm outflow. *Journal of Applied Meteorology*, 1996. doi: 10.1175/1520-0450(1996)035<1028:EORTDT>2.0.CO;2. URL https://doi.org/10.1175/1520-0450(1996)035<1028:eortdt>2.0.co;2.

Andrew Brown, Andrew Dowdy, Todd P. Lane, and Stacey Hitchcock. Types of severe convective wind events in eastern australia. **Monthly Weather Review* 151, 419–448*, 2023. doi: 10.1175/MWR-D-22-0096.1. URL https://journals.ametsoc.org/view/journals/mwre/151/2/MWR-D-22-0096.1.xml.

Federico Canepa, Massimiliano Burlando, and Giovanni Solari. Vertical profile characteristics of thunderstorm outflows. *Journal of Wind Engineering and Industrial Aerodynamics*, 2020. doi: 10.1016/j.jweia.2020.104332. URL https://www.sciencedirect.com/science/article/pii/S0167610520302427.

Guangzhao Chen and Franklin T. Lombardo. An automated classification method of thunderstorm and non-thunderstorm wind data based on a convolutional neural network. *Journal of Wind Engineering and Industrial Aerodynamics*, 2020. doi: 10.1016/j.jweia.2020.104407. URL https://doi.org/10.1016/j.jweia.2020.104407.

Samuel J. Childs, Russ S. Schumacher, and Rebecca D. Adams-Selin. High-resolution observations of a destructive macroburst. 2021. doi: 10.1175/MWR-D-20-0412.1. URL https://journals.ametsoc.org/view/journals/mwre/149/9/MWR-D-20-0412.1.xml.

John L. Cintineo, Michael J. Pavolonis, and Justin M. Sieglaff. Probsevere version 3: Improved exploitation of data fusion and machine learning for nowcasting severe weather. *Weather*

and Forecasting, 2024. doi: 10.1175/waf-d-24-0076.1. URL https://doi.org/10.1175/waf-d-24-0076.1.

Nicholas J. Cook. Confidence limits for extreme wind speeds in mixed climates. *Journal of Wind Engineering and Industrial Aerodynamics*, 2004. doi: 10.1016/j.jweia.2003.09.037.

Nicholas J. Cook. Impact of asos real-time quality control on convective gust extremes in the usa. EngrXiv Preprint, Version 1, 2023a.

Nicholas J. Cook. Automated classification of gust events in the contiguous usa. *Journal of Wind Engineering and Industrial Aerodynamics*, 2023b. doi: 10.1016/j.jweia.2023.105330. URL https://doi.org/10.1016/j.jweia.2023.105330.

Nicholas John Cook. Extreme convective gusts in the contiguous usa. *Meteorology*, 2024. doi: 10.3390/meteorology3030015. URL https://doi.org/10.3390/meteorology3030015.

Roger Edwards, John T. Allen, and Gregory W. Carbin. Reliability and climatological impacts of convective wind estimations. *Journal of Applied Meteorology and Climatology*, 2018. doi: 10.1175/JAMC-D-17-0306.1. URL https://journals.ametsoc.org/view/journals/apme/57/8/jamc-d-17-0306.1.xml.

Fanfu Fan and Weichiang Pang. Stochastic track model for tornado risk assessment in the u.s. *Frontiers in Built Environment*, 2019. doi: 10.3389/fbuil.2019.00037. URL https://doi.org/10.3389/fbuil.2019.00037.

T. Theodore Fujita. The downburst, microburst and macroburst, 1985.

Patrizia De Gaetano, Maria Pia Repetto, Teresa Repetto, and Giovanni Solari. Separation and classification of extreme wind events from anemometric records. *Journal of Wind Engineering and Industrial Aerodynamics*, 2014. doi: 10.1016/j.jweia.2014.01.006. URL https://doi.org/10.1016/j.jweia.2014.01.006.

Thomas L. Gard, Henry E. Fuelberg, and John L. Cintineo. The utility of probsevere v2.0 for predicting pulse severe thunderstorms. *Weather and Forecasting*, 2022. doi: 10.1175/waf-d-22-0003.1. URL https://doi.org/10.1175/waf-d-22-0003.1.

Maci N. Gibson and Jacob T. Carlin. Analysis of polarimetric radar downburst precursors using automated storm identification and tracking. *Weather and Forecasting*, 40, 2025. doi: 10.1175/WAF-D-24-0222.1.

L. Gomes and B. J. Vickery. Extreme wind speeds in mixed wind climates. *Journal of Wind Engineering and Industrial Aerodynamics*, 1978. doi: 10.1016/0167-6105(78)90018-1. URL https://www.sciencedirect.com/science/article/pii/0167610578900181.

W. Scott Gunter and Quint M. Long. A mesonet-based climatology of severe convective winds in west texas. *Electronic Journal of Severe Storms Meteorology*, 19(1):1–30, 2024. doi: 10.55599/ejssm.v19i1.92.

W. Scott Gunter and John L. Schroeder. High-resolution full-scale measurements of thunderstorm outflow winds. *Journal of Wind Engineering and Industrial Aerodynamics*, 2015. doi: 10.1016/j.jweia.2014.12.005. URL https://www.sciencedirect.com/science/article/pii/S0167610514002633.

William Scott Gunter. Thunderstorm outflow winds as measured by the ttuka mobile doppler radars and sticknet with applications to wind engineering. *Texas Tech University PhD dissertation*, 2015. URL http://hdl.handle.net/2346/73716.

Mohammad Hadavi and Djordje Romanic. Machine learning investigation of downburst-prone environments in canada. 2024a. doi: 10.1175/JAMC-D-23-0178.1. URL https://journals.ametsoc.org/view/journals/apme/63/6/JAMC-D-23-0178.1.xml.

Mohammad Hadavi and Djordje Romanic. Atmospheric conditions conducive to thunderstorms with downbursts in canada and a downburst precursor parameter. *Atmospheric Research*, 2024b. doi: 10.1016/j.atmosres.2024.107428. URL https://www.sciencedirect.com/science/article/pii/S0169809524002102.

R. Ian Harris and Nicholas J. Cook. The parent wind speed distribution: Why weibull? *Journal of Wind Engineering and Industrial Aerodynamics*, 2014. doi: 10.1016/j.jweia.2014.05.005. URL https://doi.org/10.1016/j.jweia.2014.05.005.

Yinghou He, Jiahan Cai, Renjie Wang, Xiaojing He, Pakwai Chan, and Jiyang Fu. Observation of downburst wind characteristics using the doppler profiler and near-ground measurements. *Natural Hazards*, 2024. doi: 10.1007/s11069-024-06410-w. URL https://link.springer.com/10.1007/s11069-024-06410-w.

Hisato Iwashita and Fumiaki Kobayashi. Transition of meteorological variables while downburst occurrence by a high density ground surface observation network. *Journal of Wind Engineering and Industrial Aerodynamics*, 2019. doi: 10.1016/j.jweia.2018.10.007. URL https://www.sciencedirect.com/science/article/pii/S0167610518304641.

Robert H. Johns and William D. Hirt. Derechos: Widespread convectively induced windstorms. *Weather and Forecasting*, 1987. doi: 10.1175/1520-0434(1987)002<0032:DWCIW>2.0.CO;2. URL https://doi.org/10.1175/1520-0434(1987)002<0032:dwciw>2.0.co;2.

J. T. Johnson, Pamela L. MacKeen, Arthur Witt, E. De Wayne Mitchell, Gregory J. Stumpf, Michael D. Eilts, and Kevin W. Thomas. The storm cell identification and tracking algorithm: An enhanced wsr-88d algorithm. *Weather and Forecasting*, 1998. doi: 10.1175/1520-0434(1998)013<0263:tsciat>2.0.co;2. URL https://doi.org/10.1175/1520-0434(1998)013<0263:tsciat>2.0.co;2.

Dae Kun Kwon and Ahsan Kareem. Towards codification of thunderstorm/downburst using gust front factor: Model-based and data-driven perspectives. *Engineering Structures*, 2019. doi: 10.1016/j.engstruct.2019.109608. URL http://www.sciencedirect.com/science/article/pii/S0141029619306315.

Franklin T. Lombardo and Alexander S. Zickar. Characteristics of measured extreme thunderstorm near-surface wind gusts in the united states. *Journal of Wind Engineering and Industrial Aerodynamics*, 2019. doi: 10.1016/j.jweia.2019.103961. URL http://www.sciencedirect.com/science/article/pii/S0167610519305525.

Franklin T. Lombardo, Alexander S. Zickar, Horia Hangan, and Ahsan Kareem. Prediction of non-synoptic wind speeds, 2021. URL https://doi.org/10.1093/oxfordhb/9780190670252.013.7.

Susanna Mohr, Michael Kunz, Alexandra Richter, and Bodo Ruck. Statistical characteristics of convective wind gusts in germany. *Natural Hazards and Earth System Sciences*, 2017. doi: 10.5194/nhess-17-957-2017. URL https://nhess.copernicus.org/articles/17/957/2017/.

Andrew Moore. Investigating the near-surface wind fields of downbursts using a series of high-resolution idealized simulations. *Weather and Forecasting*, 39(7):1065–1086, 2024. doi: 10.1175/WAF-D-23-0164.1.

Kirsten D. Orwig and John L. Schroeder. Near-surface wind characteristics of extreme thunderstorm outflows. *Journal of Wind Engineering and Industrial Aerodynamics*, 2007. doi: 10.1016/j.jweia.2006.12.002. URL https://www.sciencedirect.com/science/article/pii/S0167610506002017.

Connor Pearson, Tian-You Yu, David Bodine, Sebastian Torres, and Anthony Reinhart. A framework for comparisons of downburst precursor observations using an all-digital phased-array weather radar. *Journal of Atmospheric and Oceanic Technology*, 2023. doi: 10.1175/jtech-d-22-0130.1. URL https://doi.org/10.1175/jtech-d-22-0130.1.

Fred H. Proctor. Numerical simulations of an isolated microburst. part i: Dynamics and structure. *Journal of the Atmospheric Sciences 45(21), 3137–3160*, 1988. URL https://journals.ametsoc.org/view/journals/atsc/45/21/1520-0469_1988_045_3137_nsoaim_2_0_co_2.xml.

Fred H. Proctor. Numerical simulations of an isolated microburst. part ii: Sensitivity experiments. *Journal of the Atmospheric Sciences 46(14), 2143–2165*, 1989. URL https://journals.ametsoc.org/view/journals/atsc/46/14/1520-0469_1989_046_2143_nsoaim_2_0_co_2.xml.

Djordje Romanic. Deterministic and constrained stochastic models of thunderstorm winds. *Journal of Wind Engineering and Industrial Aerodynamics*, 2024. doi: 10.1016/j.jweia.2024.105851. URL https://www.sciencedirect.com/science/article/pii/S0167610524002149.

Djordje Romanic and Horia Hangan. Experimental investigation of the interaction between near-surface atmospheric boundary layer winds and downburst outflows. *Journal of Wind Engineering and Industrial Aerodynamics*, 2020. doi: 10.1016/j.jweia.2020.104323. URL https://www.sciencedirect.com/science/article/pii/S0167610520302336.

Djordje Romanic, Junayed Chowdhury, Jubayer Chowdhury, and Horia Hangan. Investigation of the transient nature of thunderstorm winds from europe, the united states, and australia using a new method for detection of changepoints in wind speed records. *Monthly Weather Review 148, 3747–3771*, 2020. doi: 10.1175/MWR-D-19-0312.1. URL https://journals.ametsoc.org/view/journals/mwre/148/9/mwr-d-19-0312.1.xml.

Djordje Romanic, Mateusz Taszarek, and Harold Brooks. Convective environments leading to microburst, macroburst and downburst events across the united states. *Weather and Climate Extremes*, 2022. doi: 10.1016/j.wace.2022.100474. URL https://www.sciencedirect.com/science/article/pii/S2212094722000561.

Richard Rotunno, Joseph B. Klemp, and Morris L. Weisman. A theory for strong, long-lived squall lines. *Journal of the Atmospheric Sciences*, 1988. doi: 10.1175/1520-0469(1988)045<0463:

ATFSLL>2.0.CO;2. URL https://doi.org/10.1175/1520-0469(1988)045<0463:atfsll>2.0.co;2.

Russ S. Schumacher, Samuel J. Childs, and Rebecca D. Adams-Selin. Intense surface winds from gravity wave breaking in simulations of a destructive macroburst. 2023. doi: 10.1175/MWR-D-22-0103.1. URL https://journals.ametsoc.org/view/journals/mwre/151/3/MWR-D-22-0103.1.xml.

Bryan T. Smith, Richard L. Thompson, Jeremy S. Grams, Chris Broyles, and Harold E. Brooks. Convective modes for significant severe thunderstorms in the contiguous united states. part i: Storm classification and climatology. *Weather and Forecasting 27, 1114–1135*, 2012. doi: 10.1175/WAF-D-11-00115.1. URL https://journals.ametsoc.org/view/journals/wefo/27/5/waf-d-11-00115_1.xml.

Giovanni Solari, Massimiliano Burlando, Patrizia De Gaetano, and Maria Pia Repetto. Characteristics of thunderstorms relevant to the wind loading of structures. *Wind and Structures*, 2015. doi: 10.12989/WAS.2015.20.6.763. URL https://doi.org/10.12989/WAS.2015.20.6.763.

Xinxin Sui, John Nielsen-Gammon, Zong-Liang Yang, and Dev Niyogi. Divergent urban storm response to convective, frontal and tropical systems. *Nature*, 2026. doi: 10.1038/s41586-026-10479-7. URL https://doi.org/10.1038/s41586-026-10479-7.

Robert J. Trapp, Eric D. Robinson, Michael E. Baldwin, Noah S. Diffenbaugh, and Benjamin R. J. Schwedler. Regional climate of hazardous convective weather through high-resolution dynamical downscaling. *Climate Dynamics*, 2010. doi: 10.1007/s00382-010-0826-y. URL https://doi.org/10.1007/s00382-010-0826-y.

Matthew B. Vallis, Acir Mércio Loredo-Souza, Vanessa Ferreira, and Ernani de Lima Nascimento. Classification and identification of synoptic and non-synoptic extreme wind events from surface observations in south america. *Journal of Wind Engineering and Industrial Aerodynamics*, 2019. doi: 10.1016/j.jweia.2019.103963. URL https://www.sciencedirect.com/science/article/pii/S0167610519305549.

Roger M. Wakimoto. The life cycle of thunderstorm gust fronts as viewed with doppler radar and rawinsonde data. *Monthly Weather Review*, 1982. doi: 10.1175/1520-0493(1982)110<1060:TLCOTG>2.0.CO;2. URL https://doi.org/10.1175/1520-0493(1982)110<1060:tlcotg>2.0.co;2.

Morris L. Weisman. The role of convectively generated rear-inflow jets in the evolution of long-lived mesoconvective systems. *Journal of the Atmospheric Sciences*, 1992. doi: 10.1175/1520-0469(1992)049<1826:TROCGR>2.0.CO;2. URL https://doi.org/10.1175/1520-0469(1992)049<1826:trocgr>2.0.co;2.

Arthur Witt, Michael D. Eilts, Gregory J. Stumpf, E. De Wayne Mitchell, J. T. Johnson, and Kevin W. Thomas. Evaluating the performance of wsr-88d severe storm detection algorithms. *Weather and Forecasting*, 1998. doi: 10.1175/1520-0434(1998)013<0513:etpows>2.0.co;2. URL https://doi.org/10.1175/1520-0434(1998)013<0513:etpows>2.0.co;2.

Teng Wu, Reda Snaiki, Frank Lombardo, Sihan Li, Michele Barbato, Guirong Yan, and Robert J. Trapp. Impact of climate change on extreme windstorms and implications for structural design in the united states. *Journal of Structural Engineering*, 2026. doi: 10.1061/JSENDH.STENG-15800. URL https://doi.org/10.1061/JSENDH.STENG-15800.

Shi Zhang, Giovanni Solari, Patrizia De Gaetano, Massimiliano Burlando, and Maria Pia Repetto. A refined analysis of thunderstorm outflow characteristics relevant to the wind loading of structures. *Probabilistic Engineering Mechanics*, 2018a. doi: 10.1016/j.probengmech.2017.06.003. URL http://www.sciencedirect.com/science/article/pii/S0266892017301303.

Shi Zhang, Giovanni Solari, Qingshan Yang, and Maria Pia Repetto. Extreme wind speed distribution in a mixed wind climate. *Journal of Wind Engineering and Industrial Aerodynamics*, 2018b. doi: 10.1016/j.jweia.2018.03.019. URL https://www.sciencedirect.com/science/article/pii/S0167610517309066.